STA302 Final Project

```r
library(dplyr)
```

```
## Warning: package 'dplyr' was built under R version 4.4.3

##
## Attaching package: 'dplyr'

## The following objects are masked from 'package:stats':
##
##     filter, lag

## The following objects are masked from 'package:base':
##
##     intersect, setdiff, setequal, union
```

```r
# 读入 Part 1 清洗好的数据（放在当前项目文件夹）

df <- read.csv("ames_final_cleaned.csv")

# 如果还没有 Age_Remodel，就用 Year_Sold - Year_Remod_Add 创建

if (!("Age_Remodel" %in% names(df)) &&
all(c("Year_Sold", "Year_Remod_Add") %in% names(df))) {
df$Age_Remodel <- df$Year_Sold - df$Year_Remod_Add
}

# 和 Part 1 一样，把 Age_Remodel < 0 的点删掉

if ("Age_Remodel" %in% names(df)) {
df <- df[df$Age_Remodel >= 0, ]
}

dim(df)
```

```
## [1] 2927   10
```

```r
head(df[, c("Sale_Price","Age_Remodel")])
```

```
##   Sale_Price Age_Remodel
## 1     215000          50
## 2     105000          49
## 3     172000          52
## 4     244000          42
## 5     189900          12
## 6     195500          12
```

```r
set.seed(302)

n <- nrow(df)
train_prop <- 0.8
train_idx  <- sample(seq_len(n), size = floor(train_prop * n))

train <- df[train_idx, ]
test  <- df[-train_idx, ]

dim(train); dim(test)
```

```
## [1] 2341   10
```

```
## [1] 586  10
```

```r
# 在 train / test 中创建 Bldg_Type2（OneFam vs Non1Fam）

train <- train %>%
mutate(Bldg_Type2 = ifelse(as.character(Bldg_Type) == "OneFam",
"OneFam", "Non1Fam"),
Bldg_Type2 = factor(Bldg_Type2, levels = c("OneFam","Non1Fam")))

test <- test %>%
mutate(Bldg_Type2 = ifelse(as.character(Bldg_Type) == "OneFam",
"OneFam", "Non1Fam"),
Bldg_Type2 = factor(Bldg_Type2, levels = c("OneFam","Non1Fam")))

# 只用 train 数据拟合原始 Sale_Price 模型

model_train <- lm(
Sale_Price ~ Gr_Liv_Area + Mas_Vnr_Area + Age_Remodel +
Bedroom_AbvGr + Total_Bsmt_SF + Garage_Cars +
Full_Bath + Fireplaces + Bldg_Type2 +
Gr_Liv_Area:Bldg_Type2,
data = train
)

summary(model_train)
```

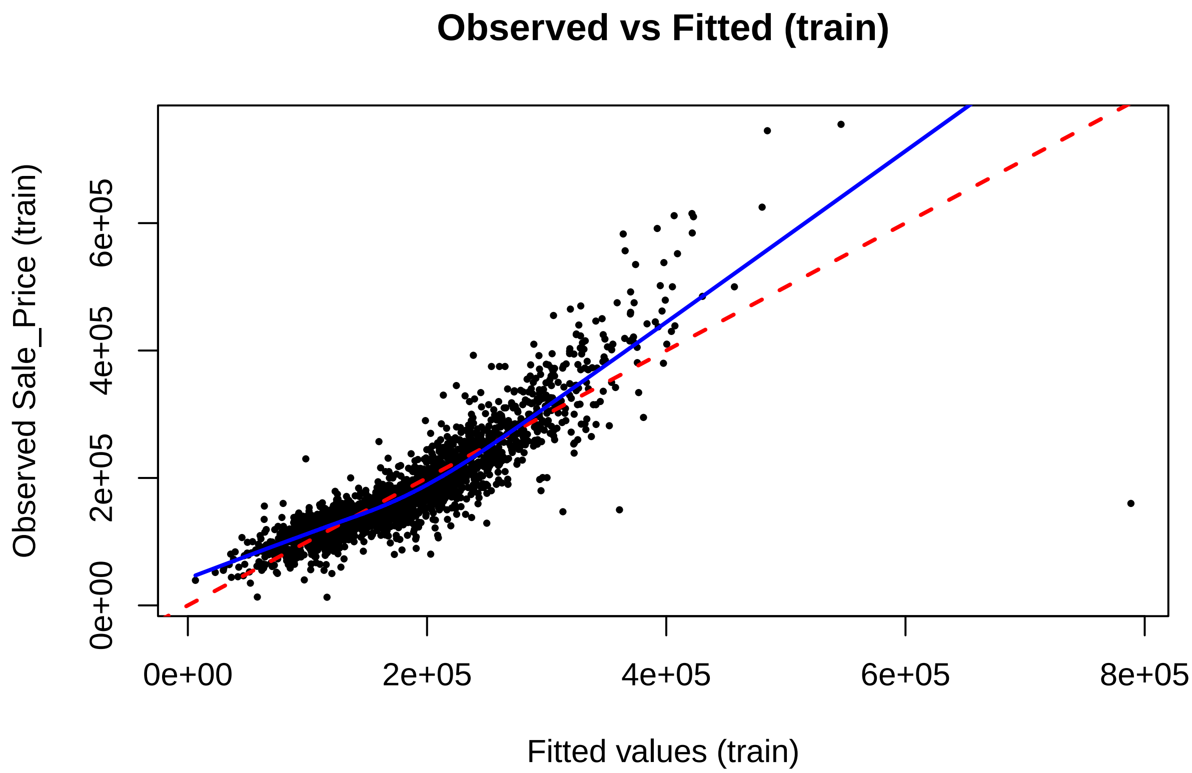
```
##
## Call:
## lm(formula = Sale_Price ~ Gr_Liv_Area + Mas_Vnr_Area + Age_Remodel +
##     Bedroom_AbvGr + Total_Bsmt_SF + Garage_Cars + Full_Bath +
##     Fireplaces + Bldg_Type2 + Gr_Liv_Area:Bldg_Type2, data = train)
##
## Residuals:
##     Min      1Q  Median      3Q     Max
## -628482  -19269   -1296   16933  260473
##
## Coefficients:
##                                  Estimate Std. Error t value Pr(>|t|)
## (Intercept)                     33382.994   4457.583   7.489 9.79e-14 ***
```

```
## Gr_Liv_Area                       70.468      2.678  26.314  < 2e-16 ***
## Mas_Vnr_Area                      61.090      4.897  12.474  < 2e-16 ***
## Age_Remodel                     -765.266     44.392 -17.239  < 2e-16 ***
## Bedroom_AbvGr                 -12020.053   1241.398  -9.683  < 2e-16 ***
## Total_Bsmt_SF                     44.866      2.135  21.017  < 2e-16 ***
## Garage_Cars                    17450.569   1312.533  13.295  < 2e-16 ***
## Full_Bath                       5058.687   2023.217   2.500  0.01248 *
## Fireplaces                      8601.324   1402.459   6.133 1.01e-09 ***
## Bldg_Type2Non1Fam              20534.627   7589.219   2.706  0.00686 **
## Gr_Liv_Area:Bldg_Type2Non1Fam    -26.850      5.047  -5.320 1.14e-07 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 37080 on 2330 degrees of freedom
## Multiple R-squared:  0.7912, Adjusted R-squared:  0.7903
## F-statistic: 882.6 on 10 and 2330 DF,  p-value: < 2.2e-16
```

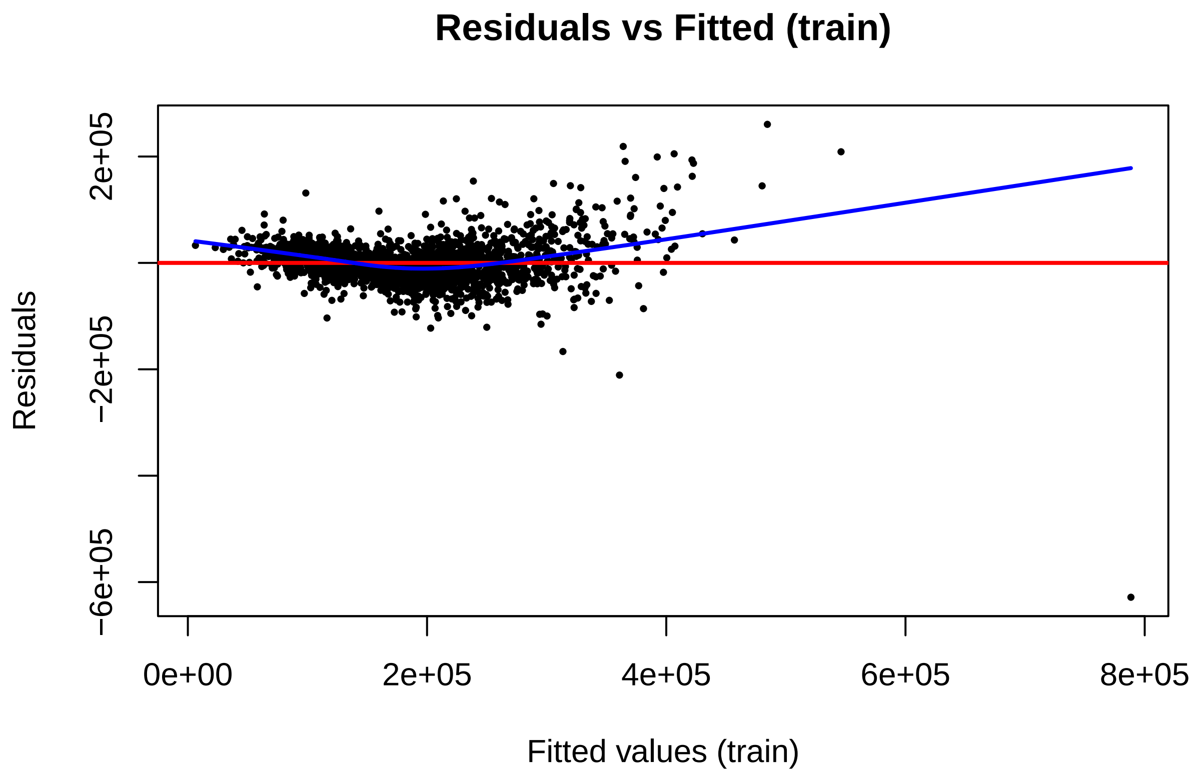
```r
y_obs_train <- model.response(model.frame(model_train))
y_hat_train <- fitted(model_train)
resid_train <- resid(model_train)

c(length(y_obs_train), length(y_hat_train), length(resid_train))
```

```
## [1] 2341 2341 2341
```

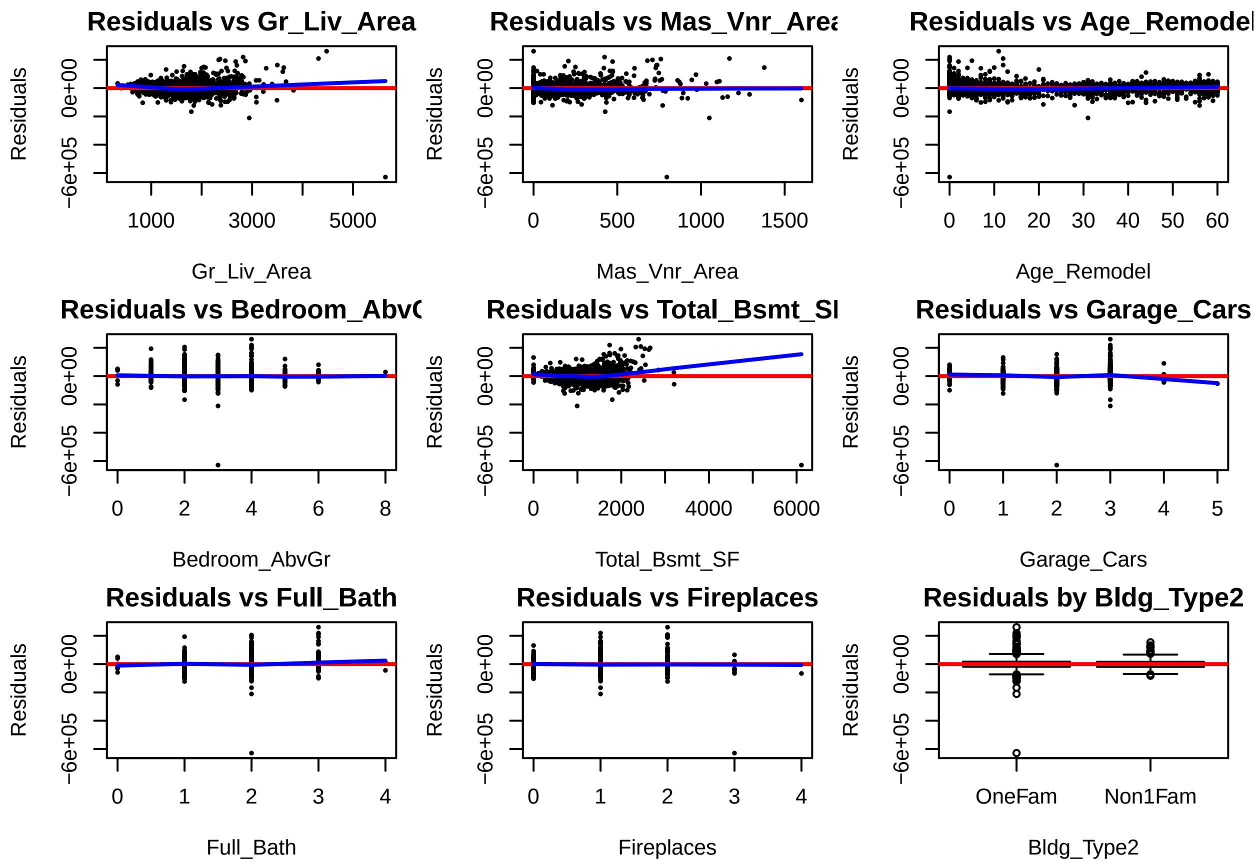
```r
plot(y_hat_train, y_obs_train,
pch = 16, cex = 0.5,
xlab = "Fitted values (train)",
ylab = "Observed Sale_Price (train)",
main = "Observed vs Fitted (train)")
abline(0, 1, col = "red", lwd = 2, lty = 2)              # 完美拟合线
lines(lowess(y_hat_train, y_obs_train), col = "blue", lwd = 2)  # LOESS 曲线
```




```r
plot(y_hat_train, resid_train,
pch = 16, cex = 0.5,
xlab = "Fitted values (train)",
ylab = "Residuals",
main = "Residuals vs Fitted (train)")
abline(h = 0, col = "red", lwd = 2)
lines(lowess(y_hat_train, resid_train), col = "blue", lwd = 2)
```

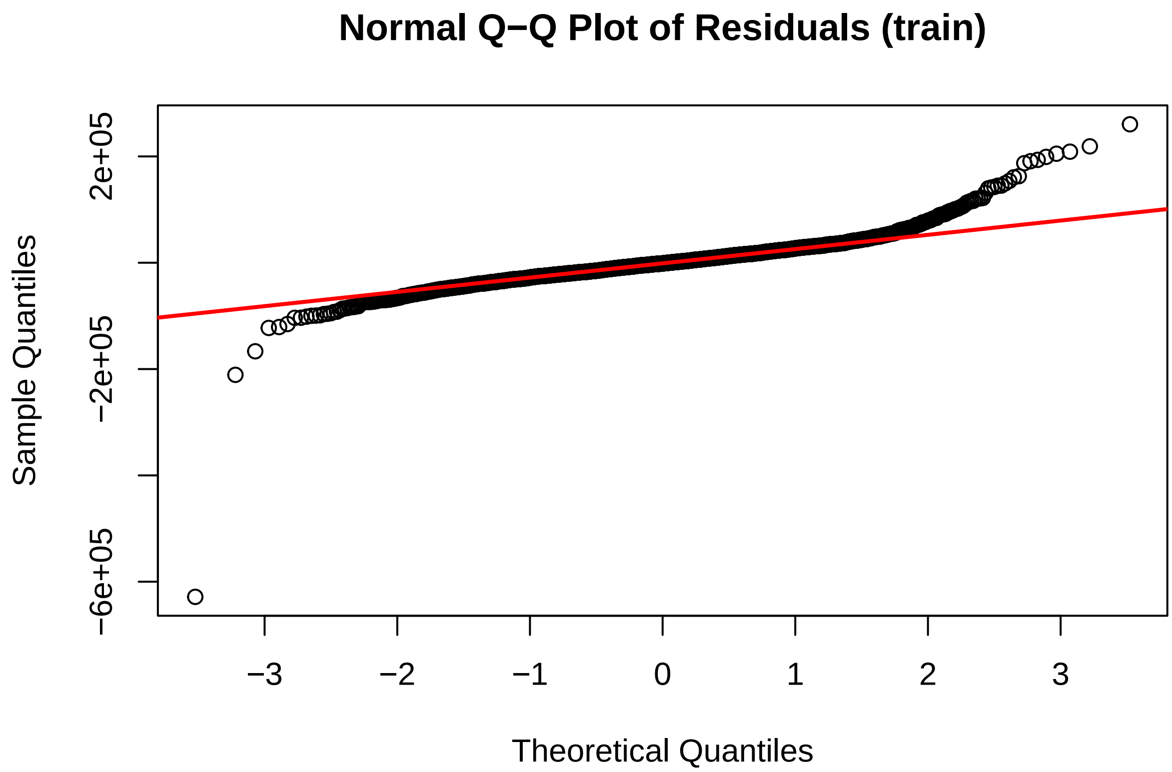



```r
num_x <- c("Gr_Liv_Area","Mas_Vnr_Area","Age_Remodel","Bedroom_AbvGr",
"Total_Bsmt_SF","Garage_Cars","Full_Bath","Fireplaces")

par(mfrow = c(3,3), mar = c(4,4,2,1))
for (v in num_x) {
x <- train[[v]]
plot(x, resid_train,
pch = 16, cex = 0.5,
xlab = v, ylab = "Residuals",
main = paste("Residuals vs", v))
abline(h = 0, col = "red", lwd = 2)
lines(lowess(x, resid_train), col = "blue", lwd = 2)
}

# 第 9 张图：分类变量 Bldg_Type2 的残差箱线图

boxplot(resid_train ~ train$Bldg_Type2,
xlab = "Bldg_Type2", ylab = "Residuals",
main = "Residuals by Bldg_Type2")
abline(h = 0, col = "red", lwd = 2)
```

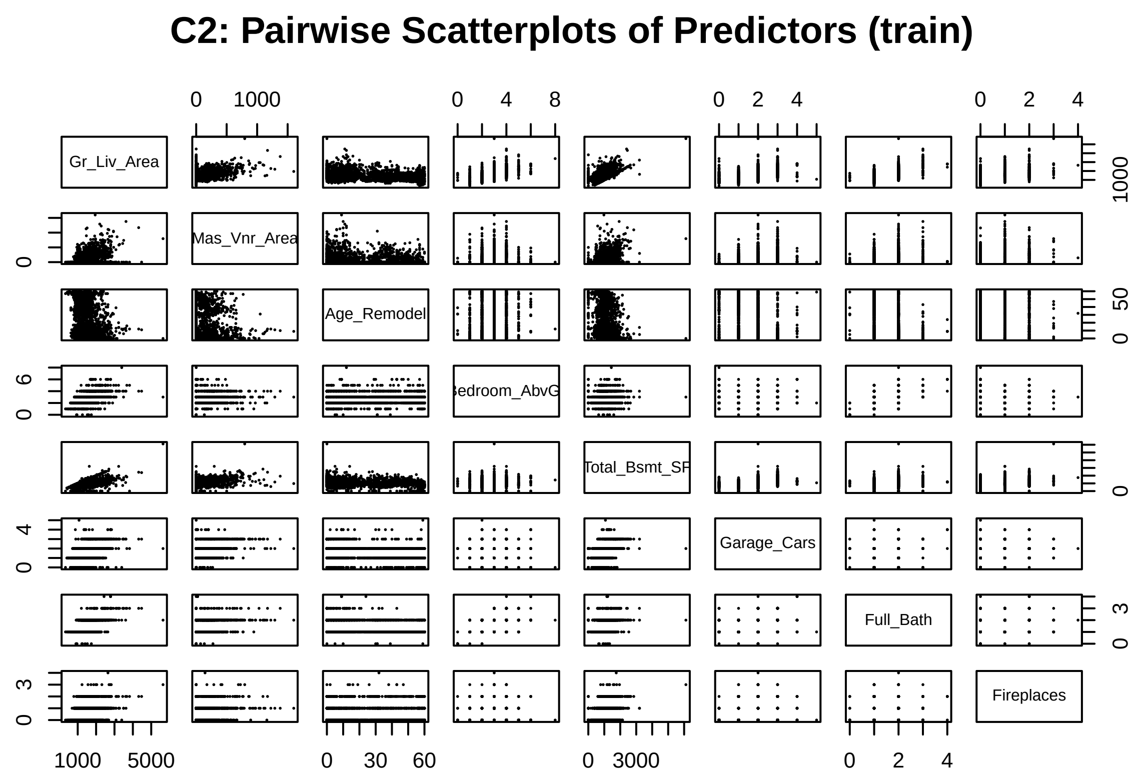



```r
par(mfrow = c(1,1))
```

```r
qqnorm(resid_train,
main = "Normal Q-Q Plot of Residuals (train)")
qqline(resid_train, col = "red", lwd = 2)
```

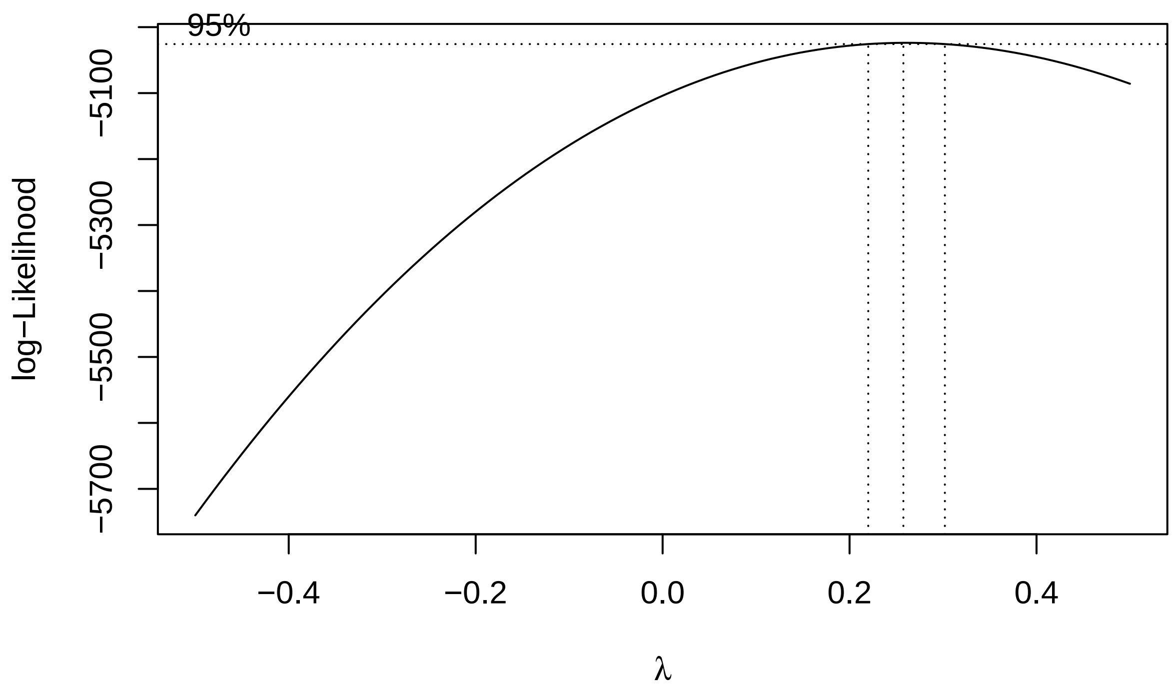



```
C2_data <- train[, num_x]

pairs(C2_data,
pch = 16, cex = 0.3,
main = "C2: Pairwise Scatterplots of Predictors (train)")
```



Chunk 1：Box–Cox 找 ＋ 保存 Y_bc 和 lambda_list

```
library(MASS)

## 4.1 对 Y 做 Box-Cox
bc <- boxcox(model_train, lambda = seq(-0.5, 0.5, by = 0.05))
```




```
lambda_y_hat <- bc$x[which.max(bc$y)]    # 真正的最优 （写在报告里）
lambda_y_hat
```

```
## [1] 0.2575758
```

```
lambda_y <- 0.25                           # 模型里实际使用的 （四次方根）

# Box-Cox 通用变换函数
bc_transform <- function(x, lambda) {
  if (abs(lambda) < 1e-6) {
    log(x)
  } else {
    (x^lambda - 1) / lambda
  }
}

# 生成变换后的响应变量 Y_bc
train$Y_bc <- bc_transform(train$Sale_Price, lambda_y)
y_bc <- train$Y_bc    # 方便后面 lm 使用

## 4.2 对每个 X 单独做 Box-Cox，先记录 到 lambda_list（用于决定怎么近似）
predictor_names <- c("Gr_Liv_Area", "Total_Bsmt_SF", "Mas_Vnr_Area",
                     "Age_Remodel", "Bedroom_AbvGr", "Garage_Cars",
                     "Full_Bath", "Fireplaces")
```

```r
lambda_list <- numeric(length(predictor_names))
names(lambda_list) <- predictor_names

for (v in predictor_names) {

  x <- train[[v]]

  # Box-Cox 要求 >0，有 0 或负数就整体 +1（只在变换时用）
  if (any(x <= 0)) {
    x_use <- x + 1
  } else {
    x_use <- x
  }

  # 对单个 predictor 做 Box-Cox: Y_bc ~ x_use
  bc_x <- boxcox(
    lm(y_bc ~ x_use),
    lambda = seq(-0.5, 0.5, by = 0.05),
    plotit = FALSE
  )

  best_lambda <- bc_x$x[which.max(bc_x$y)]
  lambda_list[v] <- best_lambda
}

lambda_list
```

```
##   Gr_Liv_Area Total_Bsmt_SF  Mas_Vnr_Area   Age_Remodel Bedroom_AbvGr
##          0.50          0.50          0.50          0.05          0.10
##   Garage_Cars     Full_Bath    Fireplaces
##          0.35          0.25          0.25
```

```r
## 手动根据 lambda_list 做变换（可解释性更好）

# 变换规则:
#     0.50 → sqrt()
#     0.05 → log()
#     0.35 → 立方根（1/3 次方）
#     0.25 → 四次方根（1/4 次方）

### -------- 1)    0.50  → sqrt() --------
train$Gr_Liv_Area_bc    <- sqrt(train$Gr_Liv_Area)
train$Total_Bsmt_SF_bc  <- sqrt(train$Total_Bsmt_SF)
train$Mas_Vnr_Area_bc   <- sqrt(train$Mas_Vnr_Area)

### -------- 2)    0.05 → log() --------
# 防止 0 值报错，统一 +1 再 log
train$Age_Remodel_bc    <- log(train$Age_Remodel + 1)
train$Bedroom_AbvGr_bc  <- log(train$Bedroom_AbvGr + 1)

### -------- 3)    0.35 → 立方根 --------
train$Garage_Cars_bc    <- (train$Garage_Cars)^(1/3)
```

```r
### -------- 4)    0.25 → 四次方根 --------
train$Full_Bath_bc      <- (train$Full_Bath)^(1/4)
train$Fireplaces_bc     <- (train$Fireplaces)^(1/4)

## 输出检查
summary(train[, c("Gr_Liv_Area_bc","Total_Bsmt_SF_bc","Mas_Vnr_Area_bc",
                  "Age_Remodel_bc","Bedroom_AbvGr_bc","Garage_Cars_bc",
                  "Full_Bath_bc","Fireplaces_bc")])
```

```
##  Gr_Liv_Area_bc  Total_Bsmt_SF_bc Mas_Vnr_Area_bc  Age_Remodel_bc
##  Min.   :18.28   Min.   : 0.00    Min.   : 0.000   Min.   :0.000
##  1st Qu.:33.63   1st Qu.:28.20    1st Qu.: 0.000   1st Qu.:1.609
##  Median :38.16   Median :31.50    Median : 0.000   Median :2.773
##  Mean   :38.32   Mean   :31.52    Mean   : 5.961   Mean   :2.587
##  3rd Qu.:41.95   3rd Qu.:36.17    3rd Qu.:12.806   3rd Qu.:3.761
##  Max.   :75.11   Max.   :78.17    Max.   :40.000   Max.   :4.111
##  Bedroom_AbvGr_bc Garage_Cars_bc   Full_Bath_bc   Fireplaces_bc
##  Min.   :0.000    Min.   :0.000   Min.   :0.000   Min.   :0.0000
##  1st Qu.:1.099    1st Qu.:1.000   1st Qu.:1.000   1st Qu.:0.0000
##  Median :1.386    Median :1.260   Median :1.189   Median :1.0000
##  Mean   :1.328    Mean   :1.149   Mean   :1.105   Mean   :0.5339
##  3rd Qu.:1.386    3rd Qu.:1.260   3rd Qu.:1.189   3rd Qu.:1.0000
##  Max.   :2.197    Max.   :1.710   Max.   :1.414   Max.   :1.4142
```

```r
model_bc <- lm(
Y_bc ~ Gr_Liv_Area_bc + Mas_Vnr_Area_bc + Age_Remodel_bc +
Bedroom_AbvGr_bc + Total_Bsmt_SF_bc + Garage_Cars_bc +
Full_Bath_bc + Fireplaces_bc + Bldg_Type2 +
Gr_Liv_Area_bc:Bldg_Type2,
data = train
)

summary(model_bc)
```

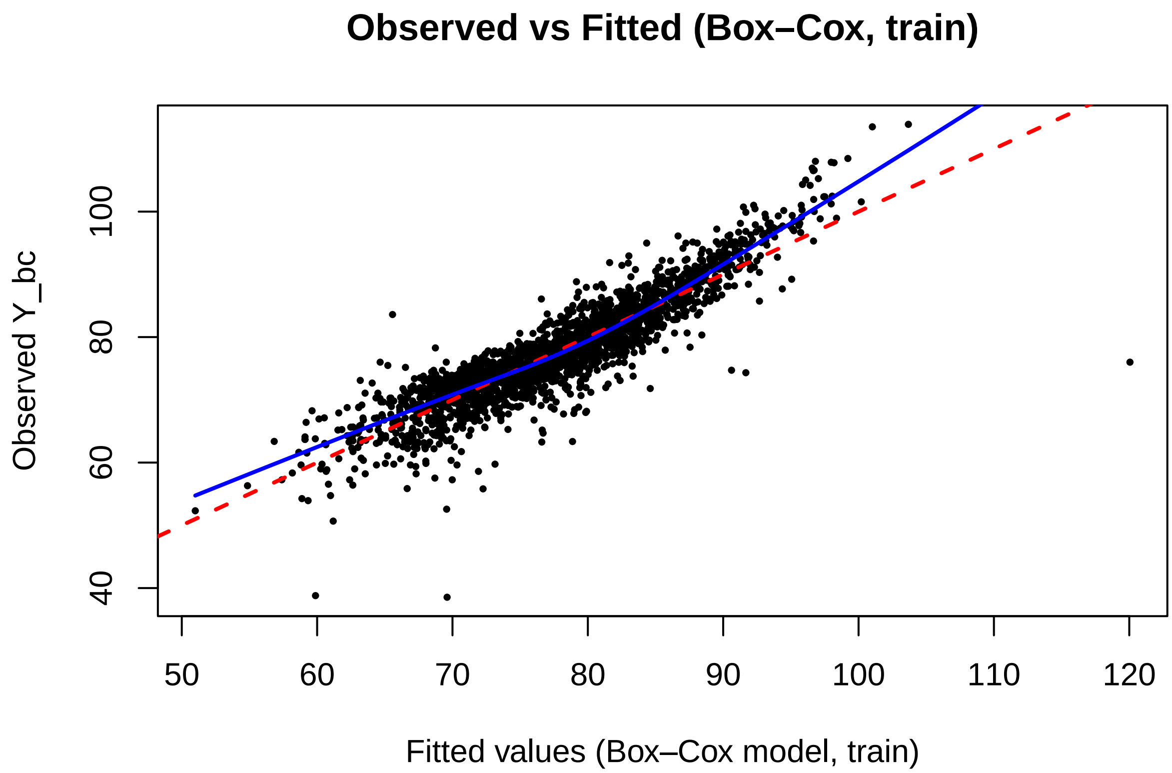
```
##
## Call:
## lm(formula = Y_bc ~ Gr_Liv_Area_bc + Mas_Vnr_Area_bc + Age_Remodel_bc +
##     Bedroom_AbvGr_bc + Total_Bsmt_SF_bc + Garage_Cars_bc + Full_Bath_bc +
##     Fireplaces_bc + Bldg_Type2 + Gr_Liv_Area_bc:Bldg_Type2, data = train)
##
## Residuals:
##     Min      1Q  Median      3Q     Max
## -44.053  -1.959   0.223   2.105  18.028
##
## Coefficients:
##                                  Estimate Std. Error t value Pr(>|t|)
## (Intercept)                      45.35987    0.94563  47.968  < 2e-16 ***
## Gr_Liv_Area_bc                    0.61675    0.01908  32.318  < 2e-16 ***
## Mas_Vnr_Area_bc                   0.10002    0.01038   9.636  < 2e-16 ***
## Age_Remodel_bc                   -1.68200    0.06714 -25.052  < 2e-16 ***
## Bedroom_AbvGr_bc                 -3.82104    0.43618  -8.760  < 2e-16 ***
## Total_Bsmt_SF_bc                  0.24299    0.01096  22.176  < 2e-16 ***
```

```
## Garage_Cars_bc                      3.95050    0.27833  14.194  < 2e-16 ***
## Full_Bath_bc                        3.84204    0.83425   4.605 4.34e-06 ***
## Fireplaces_bc                       1.74900    0.17368  10.070  < 2e-16 ***
## Bldg_Type2Non1Fam                   4.67945    1.44273   3.243   0.0012 **
## Gr_Liv_Area_bc:Bldg_Type2Non1Fam -0.16297    0.03787  -4.303 1.76e-05 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 3.607 on 2330 degrees of freedom
## Multiple R-squared:  0.815,  Adjusted R-squared:  0.8142
## F-statistic:  1026 on 10 and 2330 DF,  p-value: < 2.2e-16
```

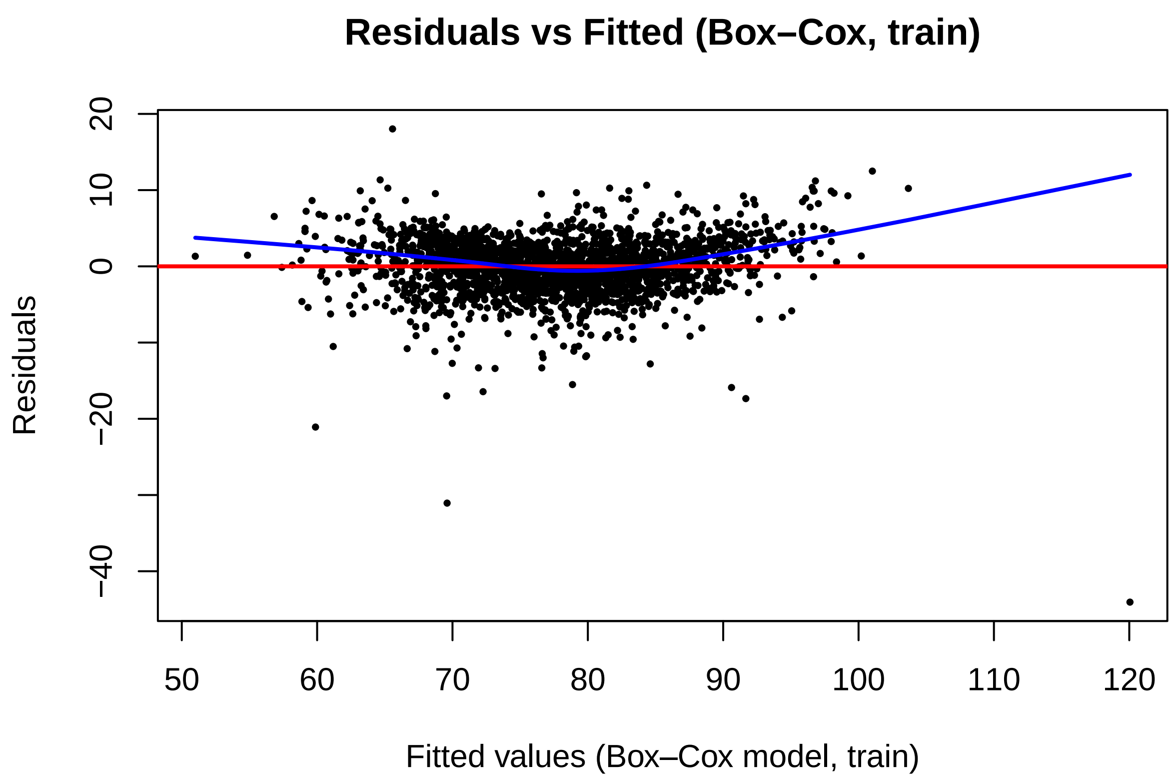
```r
y_hat_bc <- fitted(model_bc)
resid_bc <- resid(model_bc)

c(length(y_hat_bc), length(resid_bc), length(train$Y_bc))
```

```
## [1] 2341 2341 2341
```

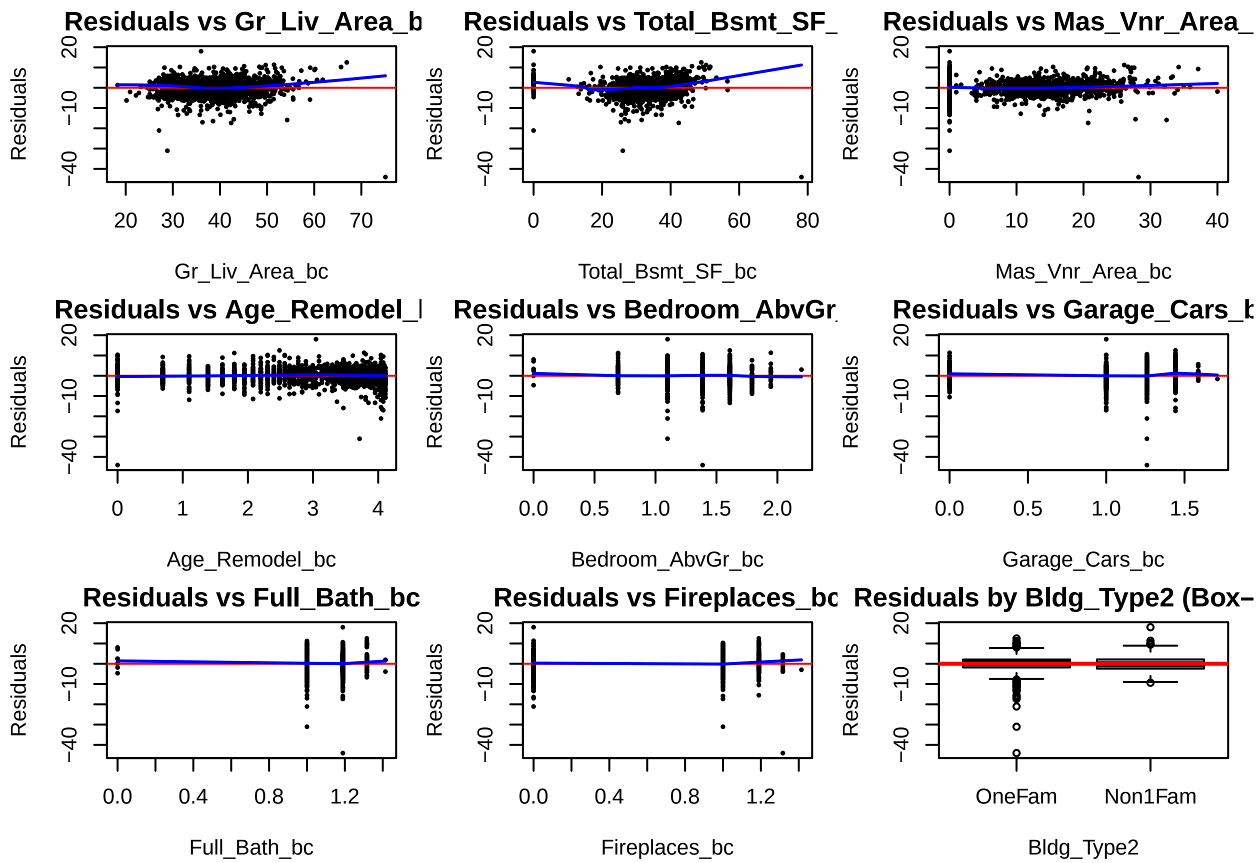
```r
y_obs_bc <- train$Y_bc

plot(y_hat_bc, y_obs_bc,
pch = 16, cex = 0.5,
xlab = "Fitted values (Box-Cox model, train)",
ylab = "Observed Y_bc",
main = "Observed vs Fitted (Box-Cox, train)")
abline(0, 1, col = "red", lwd = 2, lty = 2)
lines(lowess(y_hat_bc, y_obs_bc), col = "blue", lwd = 2)
```




```r
plot(y_hat_bc, resid_bc,
pch = 16, cex = 0.5,
xlab = "Fitted values (Box-Cox model, train)",
ylab = "Residuals",
main = "Residuals vs Fitted (Box-Cox, train)")
abline(h = 0, col = "red", lwd = 2)
lines(lowess(y_hat_bc, resid_bc), col = "blue", lwd = 2)
```

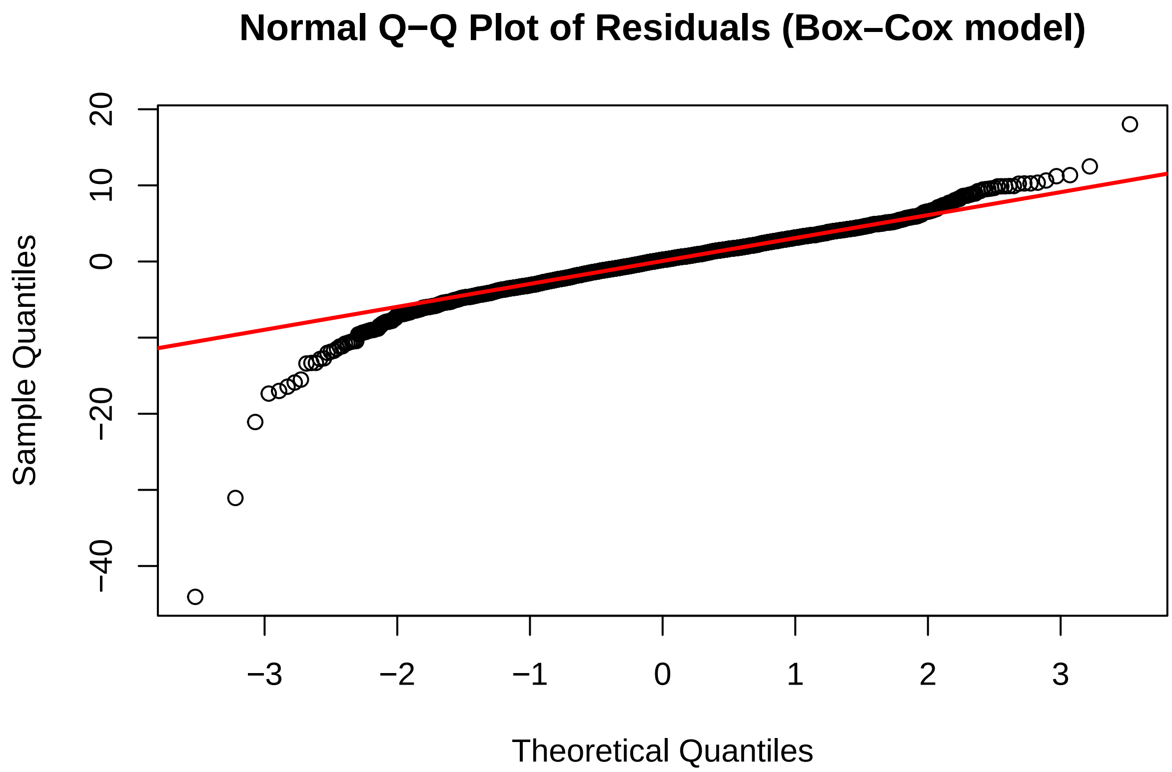



```
bc_vars <- paste0(predictor_names, "_bc")

par(mfrow = c(3,3), mar = c(4,4,2,1))

# 1-8: 数值型 X_bc

for (v in bc_vars) {
plot(train[[v]], resid_bc,
pch = 16, cex = 0.5,
xlab = v,
ylab = "Residuals",
main = paste("Residuals vs", v))
abline(h = 0, col = "red")
lines(lowess(train[[v]], resid_bc), col = "blue", lwd = 1.5)
}

# 第 9 张: Bldg_Type2

boxplot(resid_bc ~ train$Bldg_Type2,
xlab = "Bldg_Type2",
ylab = "Residuals",
main = "Residuals by Bldg_Type2 (Box-Cox)",
col = c("lightgrey","lightblue"))
abline(h = 0, col = "red", lwd = 2)
```

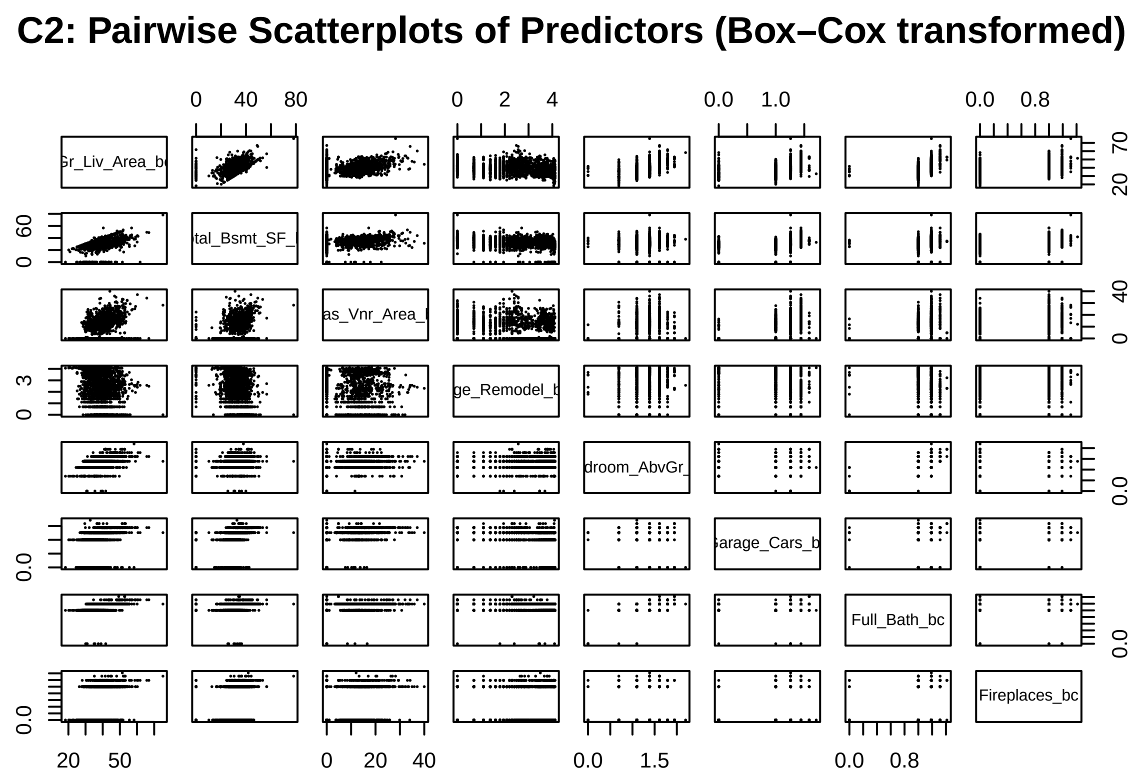



```
par(mfrow = c(1,1))
```

```
qqnorm(resid_bc,
main = "Normal Q-Q Plot of Residuals (Box-Cox model)")
qqline(resid_bc, col = "red", lwd = 2)
```




```r
bc_data <- train[, bc_vars]

pairs(bc_data,
pch = 16, cex = 0.3,
main = "C2: Pairwise Scatterplots of Predictors (Box-Cox transformed)")
```




```r
library(car)
```

```
## Warning: package 'car' was built under R version 4.4.3
```

```
## Loading required package: carData
```

```
## Warning: package 'carData' was built under R version 4.4.3
```

```
##
## Attaching package: 'car'
```

```
## The following object is masked from 'package:dplyr':
##
##     recode
```

```r
# VIF for full transformed model
vif_bc <- vif(model_bc)
```

```
## there are higher-order terms (interactions) in this model
## consider setting type = 'predictor'; see ?vif
```

```r
vif_bc
```

```
##           Gr_Liv_Area_bc           Mas_Vnr_Area_bc            Age_Remodel_bc
##                 2.576805                  1.302247                  1.360611
##          Bedroom_AbvGr_bc          Total_Bsmt_SF_bc           Garage_Cars_bc
##                 1.808096                  1.330400                  1.335042
##              Full_Bath_bc             Fireplaces_bc               Bldg_Type2
##                 1.753902                  1.453196                 53.678500
## Gr_Liv_Area_bc:Bldg_Type2
##                52.642359
```

```r
# Correlation matrix for transformed numeric predictors
bc_vars <- c("Gr_Liv_Area_bc", "Total_Bsmt_SF_bc", "Mas_Vnr_Area_bc",
             "Age_Remodel_bc", "Bedroom_AbvGr_bc", "Garage_Cars_bc",
             "Full_Bath_bc", "Fireplaces_bc")

cor_bc <- cor(train[, bc_vars])
round(cor_bc, 3)
```

```
##                  Gr_Liv_Area_bc Total_Bsmt_SF_bc Mas_Vnr_Area_bc Age_Remodel_bc
## Gr_Liv_Area_bc            1.000            0.362           0.365         -0.293
## Total_Bsmt_SF_bc          0.362            1.000           0.347         -0.292
## Mas_Vnr_Area_bc           0.365            0.347           1.000         -0.218
## Age_Remodel_bc           -0.293           -0.292          -0.218          1.000
## Bedroom_AbvGr_bc          0.503            0.049           0.081          0.051
## Garage_Cars_bc            0.362            0.315           0.294         -0.305
## Full_Bath_bc              0.560            0.230           0.235         -0.370
## Fireplaces_bc             0.470            0.294           0.292         -0.169
##                  Bedroom_AbvGr_bc Garage_Cars_bc Full_Bath_bc Fireplaces_bc
## Gr_Liv_Area_bc              0.503          0.362        0.560         0.470
## Total_Bsmt_SF_bc            0.049          0.315        0.230         0.294
## Mas_Vnr_Area_bc             0.081          0.294        0.235         0.292
## Age_Remodel_bc              0.051         -0.305       -0.370        -0.169
## Bedroom_AbvGr_bc            1.000          0.048        0.403         0.062
## Garage_Cars_bc              0.048          1.000        0.299         0.331
## Full_Bath_bc                0.403          0.299        1.000         0.233
## Fireplaces_bc               0.062          0.331        0.233         1.000
```

```r
# Overall F-test: does the model explain Y_bc better than no predictors?
mod0_bc <- lm(Y_bc ~ 1, data = train)       # null model with intercept only
anova(mod0_bc, model_bc)
```

```
## Analysis of Variance Table
##
## Model 1: Y_bc ~ 1
## Model 2: Y_bc ~ Gr_Liv_Area_bc + Mas_Vnr_Area_bc + Age_Remodel_bc + Bedroom_AbvGr_bc +
##     Total_Bsmt_SF_bc + Garage_Cars_bc + Full_Bath_bc + Fireplaces_bc +
##     Bldg_Type2 + Gr_Liv_Area_bc:Bldg_Type2
##   Res.Df    RSS Df Sum of Sq      F    Pr(>F)
## 1   2340 163853
## 2   2330  30320 10    133533 1026.2 < 2.2e-16 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

```r
summary(model_bc)
```

```
##
## Call:
## lm(formula = Y_bc ~ Gr_Liv_Area_bc + Mas_Vnr_Area_bc + Age_Remodel_bc +
##     Bedroom_AbvGr_bc + Total_Bsmt_SF_bc + Garage_Cars_bc + Full_Bath_bc +
##     Fireplaces_bc + Bldg_Type2 + Gr_Liv_Area_bc:Bldg_Type2, data = train)
##
## Residuals:
##     Min      1Q  Median      3Q     Max
## -44.053  -1.959   0.223   2.105  18.028
##
## Coefficients:
##                                Estimate Std. Error t value Pr(>|t|)
## (Intercept)                    45.35987    0.94563  47.968  < 2e-16 ***
## Gr_Liv_Area_bc                  0.61675    0.01908  32.318  < 2e-16 ***
## Mas_Vnr_Area_bc                 0.10002    0.01038   9.636  < 2e-16 ***
## Age_Remodel_bc                 -1.68200    0.06714 -25.052  < 2e-16 ***
## Bedroom_AbvGr_bc               -3.82104    0.43618  -8.760  < 2e-16 ***
## Total_Bsmt_SF_bc                0.24299    0.01096  22.176  < 2e-16 ***
## Garage_Cars_bc                  3.95050    0.27833  14.194  < 2e-16 ***
## Full_Bath_bc                    3.84204    0.83425   4.605 4.34e-06 ***
## Fireplaces_bc                   1.74900    0.17368  10.070  < 2e-16 ***
## Bldg_Type2Non1Fam               4.67945    1.44273   3.243   0.0012 **
## Gr_Liv_Area_bc:Bldg_Type2Non1Fam -0.16297    0.03787  -4.303 1.76e-05 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 3.607 on 2330 degrees of freedom
## Multiple R-squared:  0.815,  Adjusted R-squared:  0.8142
## F-statistic:  1026 on 10 and 2330 DF,  p-value: < 2.2e-16
```

```r
#Test whether interaction term is needed
model_bc_no_int <- lm(
  Y_bc ~ Gr_Liv_Area_bc + Mas_Vnr_Area_bc + Age_Remodel_bc +
    Bedroom_AbvGr_bc + Total_Bsmt_SF_bc + Garage_Cars_bc +
    Full_Bath_bc + Fireplaces_bc + Bldg_Type2,
  data = train
)

anova(model_bc_no_int, model_bc)
```

```
## Analysis of Variance Table
##
## Model 1: Y_bc ~ Gr_Liv_Area_bc + Mas_Vnr_Area_bc + Age_Remodel_bc + Bedroom_AbvGr_bc +
##     Total_Bsmt_SF_bc + Garage_Cars_bc + Full_Bath_bc + Fireplaces_bc +
##     Bldg_Type2
## Model 2: Y_bc ~ Gr_Liv_Area_bc + Mas_Vnr_Area_bc + Age_Remodel_bc + Bedroom_AbvGr_bc +
##     Total_Bsmt_SF_bc + Garage_Cars_bc + Full_Bath_bc + Fireplaces_bc +
##     Bldg_Type2 + Gr_Liv_Area_bc:Bldg_Type2
##   Res.Df   RSS Df Sum of Sq      F    Pr(>F)
## 1   2331 30561
## 2   2330 30320  1    240.94 18.515 1.755e-05 ***
```

```
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

```r
#Test dropping the "bedroom" variable, since it overlaps with size and baths.
model_bc_no_bedroom <- lm(
  Y_bc ~ Gr_Liv_Area_bc + Mas_Vnr_Area_bc + Age_Remodel_bc +
    Total_Bsmt_SF_bc + Garage_Cars_bc + Full_Bath_bc +
    Fireplaces_bc + Bldg_Type2 + Gr_Liv_Area_bc:Bldg_Type2,
  data = train
)

anova(model_bc_no_bedroom, model_bc)
```

```
## Analysis of Variance Table
##
## Model 1: Y_bc ~ Gr_Liv_Area_bc + Mas_Vnr_Area_bc + Age_Remodel_bc + Total_Bsmt_SF_bc +
##     Garage_Cars_bc + Full_Bath_bc + Fireplaces_bc + Bldg_Type2 +
##     Gr_Liv_Area_bc:Bldg_Type2
## Model 2: Y_bc ~ Gr_Liv_Area_bc + Mas_Vnr_Area_bc + Age_Remodel_bc + Bedroom_AbvGr_bc +
##     Total_Bsmt_SF_bc + Garage_Cars_bc + Full_Bath_bc + Fireplaces_bc +
##     Bldg_Type2 + Gr_Liv_Area_bc:Bldg_Type2
##   Res.Df   RSS Df Sum of Sq     F    Pr(>F)
## 1   2331 31319
## 2   2330 30320  1    998.62 76.74 < 2.2e-16 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

```r
#Test dropping the "bedroom" variable, since it overlaps with size and baths.
model_bc_no_amen <- lm(
  Y_bc ~ Gr_Liv_Area_bc + Mas_Vnr_Area_bc + Age_Remodel_bc +
    Bedroom_AbvGr_bc + Total_Bsmt_SF_bc + Bldg_Type2 +
    Gr_Liv_Area_bc:Bldg_Type2,
  data = train
)

anova(model_bc_no_amen, model_bc)
```

```
## Analysis of Variance Table
##
## Model 1: Y_bc ~ Gr_Liv_Area_bc + Mas_Vnr_Area_bc + Age_Remodel_bc + Bedroom_AbvGr_bc +
##     Total_Bsmt_SF_bc + Bldg_Type2 + Gr_Liv_Area_bc:Bldg_Type2
## Model 2: Y_bc ~ Gr_Liv_Area_bc + Mas_Vnr_Area_bc + Age_Remodel_bc + Bedroom_AbvGr_bc +
##     Total_Bsmt_SF_bc + Garage_Cars_bc + Full_Bath_bc + Fireplaces_bc +
##     Bldg_Type2 + Gr_Liv_Area_bc:Bldg_Type2
##   Res.Df   RSS Df Sum of Sq      F    Pr(>F)
## 1   2333 35406
## 2   2330 30320  3    5085.8 130.28 < 2.2e-16 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

```r
library(MASS)
```

```r
step_bc_back <- stepAIC(
  model_bc,
  direction = "backward",  # start from full and only drop
  trace = TRUE            # set TRUE if you want to see each step printed
)
```

```
## Start:  AIC=6017.85
## Y_bc ~ Gr_Liv_Area_bc + Mas_Vnr_Area_bc + Age_Remodel_bc + Bedroom_AbvGr_bc +
##     Total_Bsmt_SF_bc + Garage_Cars_bc + Full_Bath_bc + Fireplaces_bc +
##     Bldg_Type2 + Gr_Liv_Area_bc:Bldg_Type2
##
##                            Df Sum of Sq   RSS    AIC
## <none>                                  30320 6017.8
## - Gr_Liv_Area_bc:Bldg_Type2  1     240.9 30561 6034.4
## - Full_Bath_bc               1     276.0 30596 6037.1
## - Bedroom_AbvGr_bc           1     998.6 31319 6091.7
## - Mas_Vnr_Area_bc            1    1208.3 31528 6107.3
## - Fireplaces_bc              1    1319.6 31640 6115.6
## - Garage_Cars_bc             1    2621.6 32942 6210.0
## - Total_Bsmt_SF_bc           1    6399.5 36720 6464.1
## - Age_Remodel_bc             1    8166.7 38487 6574.2
```

```r
summary(step_bc_back)
```

```
##
## Call:
## lm(formula = Y_bc ~ Gr_Liv_Area_bc + Mas_Vnr_Area_bc + Age_Remodel_bc +
##     Bedroom_AbvGr_bc + Total_Bsmt_SF_bc + Garage_Cars_bc + Full_Bath_bc +
##     Fireplaces_bc + Bldg_Type2 + Gr_Liv_Area_bc:Bldg_Type2, data = train)
##
## Residuals:
##     Min      1Q  Median      3Q     Max
## -44.053  -1.959   0.223   2.105  18.028
##
## Coefficients:
##                                 Estimate Std. Error t value Pr(>|t|)
## (Intercept)                     45.35987    0.94563  47.968  < 2e-16 ***
## Gr_Liv_Area_bc                   0.61675    0.01908  32.318  < 2e-16 ***
## Mas_Vnr_Area_bc                  0.10002    0.01038   9.636  < 2e-16 ***
## Age_Remodel_bc                  -1.68200    0.06714 -25.052  < 2e-16 ***
## Bedroom_AbvGr_bc                -3.82104    0.43618  -8.760  < 2e-16 ***
## Total_Bsmt_SF_bc                 0.24299    0.01096  22.176  < 2e-16 ***
## Garage_Cars_bc                   3.95050    0.27833  14.194  < 2e-16 ***
## Full_Bath_bc                     3.84204    0.83425   4.605 4.34e-06 ***
## Fireplaces_bc                    1.74900    0.17368  10.070  < 2e-16 ***
## Bldg_Type2Non1Fam                4.67945    1.44273   3.243   0.0012 **
## Gr_Liv_Area_bc:Bldg_Type2Non1Fam -0.16297    0.03787  -4.303 1.76e-05 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 3.607 on 2330 degrees of freedom
## Multiple R-squared:  0.815,  Adjusted R-squared:  0.8142
## F-statistic:  1026 on 10 and 2330 DF,  p-value: < 2.2e-16
```

```r
step_bc_step <- stepAIC(model_bc, direction = "both", trace = FALSE)
summary(step_bc_step)
```

```
##
## Call:
## lm(formula = Y_bc ~ Gr_Liv_Area_bc + Mas_Vnr_Area_bc + Age_Remodel_bc +
##     Bedroom_AbvGr_bc + Total_Bsmt_SF_bc + Garage_Cars_bc + Full_Bath_bc +
##     Fireplaces_bc + Bldg_Type2 + Gr_Liv_Area_bc:Bldg_Type2, data = train)
##
## Residuals:
##     Min      1Q  Median      3Q     Max
## -44.053  -1.959   0.223   2.105  18.028
##
## Coefficients:
##                                Estimate Std. Error t value Pr(>|t|)
## (Intercept)                    45.35987    0.94563  47.968  < 2e-16 ***
## Gr_Liv_Area_bc                  0.61675    0.01908  32.318  < 2e-16 ***
## Mas_Vnr_Area_bc                 0.10002    0.01038   9.636  < 2e-16 ***
## Age_Remodel_bc                 -1.68200    0.06714 -25.052  < 2e-16 ***
## Bedroom_AbvGr_bc               -3.82104    0.43618  -8.760  < 2e-16 ***
## Total_Bsmt_SF_bc                0.24299    0.01096  22.176  < 2e-16 ***
## Garage_Cars_bc                  3.95050    0.27833  14.194  < 2e-16 ***
## Full_Bath_bc                    3.84204    0.83425   4.605 4.34e-06 ***
## Fireplaces_bc                   1.74900    0.17368  10.070  < 2e-16 ***
## Bldg_Type2Non1Fam               4.67945    1.44273   3.243   0.0012 **
## Gr_Liv_Area_bc:Bldg_Type2Non1Fam -0.16297    0.03787  -4.303 1.76e-05 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 3.607 on 2330 degrees of freedom
## Multiple R-squared:  0.815,  Adjusted R-squared:  0.8142
## F-statistic:  1026 on 10 and 2330 DF,  p-value: < 2.2e-16
```

```r
#BIC
n_train <- nrow(train)

step_bc_BIC <- stepAIC(
  model_bc,
  direction = "backward",
  k = log(n_train),
  trace = FALSE
)

summary(step_bc_BIC)
```

```
##
## Call:
## lm(formula = Y_bc ~ Gr_Liv_Area_bc + Mas_Vnr_Area_bc + Age_Remodel_bc +
##     Bedroom_AbvGr_bc + Total_Bsmt_SF_bc + Garage_Cars_bc + Full_Bath_bc +
##     Fireplaces_bc + Bldg_Type2 + Gr_Liv_Area_bc:Bldg_Type2, data = train)
##
## Residuals:
```

```
##     Min      1Q  Median      3Q     Max
## -44.053  -1.959   0.223   2.105  18.028
##
## Coefficients:
##                              Estimate Std. Error t value Pr(>|t|)
## (Intercept)                  45.35987    0.94563  47.968  < 2e-16 ***
## Gr_Liv_Area_bc                0.61675    0.01908  32.318  < 2e-16 ***
## Mas_Vnr_Area_bc               0.10002    0.01038   9.636  < 2e-16 ***
## Age_Remodel_bc               -1.68200    0.06714 -25.052  < 2e-16 ***
## Bedroom_AbvGr_bc             -3.82104    0.43618  -8.760  < 2e-16 ***
## Total_Bsmt_SF_bc              0.24299    0.01096  22.176  < 2e-16 ***
## Garage_Cars_bc                3.95050    0.27833  14.194  < 2e-16 ***
## Full_Bath_bc                  3.84204    0.83425   4.605 4.34e-06 ***
## Fireplaces_bc                 1.74900    0.17368  10.070  < 2e-16 ***
## Bldg_Type2Non1Fam             4.67945    1.44273   3.243   0.0012 **
## Gr_Liv_Area_bc:Bldg_Type2Non1Fam -0.16297    0.03787  -4.303 1.76e-05 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 3.607 on 2330 degrees of freedom
## Multiple R-squared:  0.815,  Adjusted R-squared:  0.8142
## F-statistic:  1026 on 10 and 2330 DF,  p-value: < 2.2e-16
```

```r
# AIC(model_bc, step_bc)
# BIC(model_bc, step_bc)
#
# c(
#   full    = summary(model_bc)$adj.r.squared,
#   stepAIC = summary(step_bc)$adj.r.squared
# )
# c(
#   full    = summary(model_bc)$adj.r.squared,
#   stepAIC = summary(step_bc)$adj.r.squared
# )

model_bc
```

```
##
## Call:
## lm(formula = Y_bc ~ Gr_Liv_Area_bc + Mas_Vnr_Area_bc + Age_Remodel_bc +
##     Bedroom_AbvGr_bc + Total_Bsmt_SF_bc + Garage_Cars_bc + Full_Bath_bc +
##     Fireplaces_bc + Bldg_Type2 + Gr_Liv_Area_bc:Bldg_Type2, data = train)
##
## Coefficients:
##                      (Intercept)                    Gr_Liv_Area_bc
##                          45.3599                            0.6167
##                  Mas_Vnr_Area_bc                    Age_Remodel_bc
##                           0.1000                           -1.6820
##                 Bedroom_AbvGr_bc                  Total_Bsmt_SF_bc
##                          -3.8210                            0.2430
##                   Garage_Cars_bc                      Full_Bath_bc
##                           3.9505                            3.8420
##                    Fireplaces_bc                 Bldg_Type2Non1Fam
##                           1.7490                            4.6794
```

```
## Gr_Liv_Area_bc:Bldg_Type2Non1Fam
##                           -0.1630
```

```
step_bc_back
```

```
##
## Call:
## lm(formula = Y_bc ~ Gr_Liv_Area_bc + Mas_Vnr_Area_bc + Age_Remodel_bc +
##     Bedroom_AbvGr_bc + Total_Bsmt_SF_bc + Garage_Cars_bc + Full_Bath_bc +
##     Fireplaces_bc + Bldg_Type2 + Gr_Liv_Area_bc:Bldg_Type2, data = train)
##
## Coefficients:
##                     (Intercept)                   Gr_Liv_Area_bc
##                         45.3599                           0.6167
##                 Mas_Vnr_Area_bc                   Age_Remodel_bc
##                          0.1000                          -1.6820
##                Bedroom_AbvGr_bc                 Total_Bsmt_SF_bc
##                         -3.8210                           0.2430
##                  Garage_Cars_bc                     Full_Bath_bc
##                          3.9505                           3.8420
##                   Fireplaces_bc                Bldg_Type2Non1Fam
##                          1.7490                           4.6794
## Gr_Liv_Area_bc:Bldg_Type2Non1Fam
##                         -0.1630
```

```
step_bc_step
```

```
##
## Call:
## lm(formula = Y_bc ~ Gr_Liv_Area_bc + Mas_Vnr_Area_bc + Age_Remodel_bc +
##     Bedroom_AbvGr_bc + Total_Bsmt_SF_bc + Garage_Cars_bc + Full_Bath_bc +
##     Fireplaces_bc + Bldg_Type2 + Gr_Liv_Area_bc:Bldg_Type2, data = train)
##
## Coefficients:
##                     (Intercept)                   Gr_Liv_Area_bc
##                         45.3599                           0.6167
##                 Mas_Vnr_Area_bc                   Age_Remodel_bc
##                          0.1000                          -1.6820
##                Bedroom_AbvGr_bc                 Total_Bsmt_SF_bc
##                         -3.8210                           0.2430
##                  Garage_Cars_bc                     Full_Bath_bc
##                          3.9505                           3.8420
##                   Fireplaces_bc                Bldg_Type2Non1Fam
##                          1.7490                           4.6794
## Gr_Liv_Area_bc:Bldg_Type2Non1Fam
##                         -0.1630
```

```
step_bc_BIC
```

```
##
## Call:
## lm(formula = Y_bc ~ Gr_Liv_Area_bc + Mas_Vnr_Area_bc + Age_Remodel_bc +
```

```
##     Bedroom_AbvGr_bc + Total_Bsmt_SF_bc + Garage_Cars_bc + Full_Bath_bc +
##     Fireplaces_bc + Bldg_Type2 + Gr_Liv_Area_bc:Bldg_Type2, data = train)
##
## Coefficients:
##                      (Intercept)                    Gr_Liv_Area_bc
##                          45.3599                            0.6167
##                   Mas_Vnr_Area_bc                   Age_Remodel_bc
##                           0.1000                           -1.6820
##                  Bedroom_AbvGr_bc                 Total_Bsmt_SF_bc
##                          -3.8210                            0.2430
##                    Garage_Cars_bc                     Full_Bath_bc
##                           3.9505                            3.8420
##                     Fireplaces_bc                 Bldg_Type2Non1Fam
##                           1.7490                            4.6794
## Gr_Liv_Area_bc:Bldg_Type2Non1Fam
##                          -0.1630
```

influential points

```r
# pick one (all same) to be the final model
final_bc <- model_bc

# sample & predictor numbers
n_bc <- nrow(train)
p_bc <- length(coef(final_bc)) - 1

# cutoffs
h_cut    <- 2 * (p_bc + 1) / n_bc
cook_cut <- qf(0.5, df1 = p_bc + 1, df2 = n_bc - p_bc - 1)
fit_cut  <- 2 * sqrt((p_bc + 1) / n_bc)
beta_cut <- 2 / sqrt(n_bc)

# diagnose variables
h_ii      <- hatvalues(final_bc)          # leverage
r_std     <- rstandard(final_bc)          # standardized residuals
cook_d    <- cooks.distance(final_bc)     # Cook's distance
dffits_i  <- dffits(final_bc)             # DFFITS
dfbetas_i <- dfbetas(final_bc)            # DFBETAS (矩阵)

# problematic points
lev_pts    <- which(h_ii > h_cut)
outliers   <- which(r_std > 4 | r_std < -4)
cook_infl  <- which(cook_d > cook_cut)
fit_infl   <- which(abs(dffits_i) > fit_cut)

cat("leverage points \n", lev_pts, "\n\n")
```

```
## leverage points
##  15 30 60 75 121 127 131 140 149 160 166 167 172 180 225 230 279 289 298 306 309 320 343 344 345 353
```

```r
cat("outliers \n", outliers, "\n\n")
```

```
## outliers
##  100 423 459 1196 1330 1849 2044 2095 2251
```

```r
cat("cook distance influential \n", cook_infl, "\n\n")
```

```
## cook distance influential
##
```

```r
cat("DFFITS influential \n", fit_infl, "\n\n")
```

```
## DFFITS influential
##  3 30 39 42 65 100 107 116 131 183 204 219 280 286 289 309 313 317 339 353 362 370 372 374 383 423 4
```

```r
# debetas
for (j in 1:ncol(dfbetas_i)) {
  cat("DFBETAS for beta", colnames(dfbetas_i)[j], ":\n")
  print(which(abs(dfbetas_i[, j]) > beta_cut))
  cat("\n")
}
```

```
## DFBETAS for beta (Intercept) :
## 2686 1336 1782 2881 1995 1915 1760  728  137  807 2677 1332 2443 2493 1560 1232
##   58   95  100  107  112  116  131  183  286  298  309  317  353  362  372  373
##  766 1294 2027 2043  182  709 1949  716 1611 1181  795 2003 1996 1170 1860 1886
##  390  398  419  423  459  513  522  526  544  553  565  585  607  608  615  630
## 2330 2664   16 1981 1307 1016  666 1539 1831 2597 2844 2828 2204 2242 1178  605
##  664  693  718  751  783  800  824  865  930  974  979  983  984  993 1023 1047
## 1557  727 2448  894 1071 1767  783 1554  910 1169  700 2254  175  768 2656  721
## 1077 1093 1114 1128 1140 1156 1160 1196 1201 1206 1224 1229 1307 1322 1333 1365
##  957  661 1303 1513  984 2069 2620 2764 1507   28 2118 1221  763 2619  809 2210
## 1378 1381 1425 1426 1440 1456 1458 1467 1470 1492 1579 1583 1585 1606 1622 1643
## 1348  291 2294  220  941 2559 1258  719 2735  164 1064 2045 1956 1285 1288 2227
## 1663 1672 1705 1719 1750 1767 1780 1796 1810 1815 1838 1839 1853 1903 1904 1913
## 1386  233  170 2721 1968 1499 2832 1194 2243 1183  434 1983  155 1772  623 1347
## 1978 2008 2044 2075 2088 2095 2102 2123 2210 2251 2269 2293 2296 2310 2318 2326
##
## DFBETAS for beta Gr_Liv_Area_bc :
## 2296 2686  557 1336 1995 1915 1760  753 2251  293  724  308  901  137 2228 2443
##   39   58   72   95  112  116  131  140  171  188  204  267  280  286  289  353
##  943 2784  766 1294 2027 2043 2103  182 1949  716 1611 1181 1319 1000 1860  457
##  370  375  390  398  419  423  436  459  522  526  544  553  575  614  615  650
## 2330 2664   16   45 2484 2840 1307 1764 2648 1060 2380  187 2036 1692 2204 1178
##  664  693  718  721  732  760  783  814  861  887  894  905  927  965  984 1023
##  214 1852  727 2448 1993  967 1767  783 1522  433 2860 1554  910  700 2254  859
## 1041 1055 1093 1114 1118 1153 1156 1160 1180 1188 1191 1196 1201 1224 1229 1242
##  428  276  367  768 1556 2656  948  957  661 1303 1642 2069 1054   28 2645 2118
## 1256 1298 1312 1322 1330 1333 1335 1378 1381 1425 1435 1456 1460 1492 1553 1579
## 2454 1997  398 1348  291  220  680  941 2559 2813 2735 1498 1064 2045 1314 2890
## 1590 1603 1621 1663 1672 1719 1741 1750 1767 1809 1810 1832 1838 1839 1842 1849
```

```
## 1956 2001 1741 1285 2227  128 2164 1386  233 2216  170 2721 1499 2332 1525  286
## 1853 1862 1884 1903 1913 1915 1917 1978 2008 2043 2044 2075 2095 2147 2188 2228
## 1183  434 1772 2408
## 2251 2269 2310 2323
##
## DFBETAS for beta Mas_Vnr_Area_bc :
##  313 2296 1638  629  138 1782 2881 1760  738  109  724   49  901 2228 2677 2364
##   11   39   42   65   76  100  107  131  164  203  204  266  280  289  309  339
##   30 2443 1198 2027 2043 1325  182  716 1181 2172 1170 1860  457   94 2664   16
##  350  353  383  419  423  434  459  526  553  606  608  615  650  692  693  718
##   45 2484  888 1307   92 1849 2380 1014 1178 2760  642  322 2448  758 1281  119
##  721  732  733  783  818  852  894 1014 1023 1036 1043 1088 1114 1123 1124 1127
## 2514 1843 1071 1767  783  422  433 1554  910 1068 2499  700 2254  392 1907  789
## 1132 1133 1140 1156 1160 1185 1188 1196 1201 1208 1216 1224 1229 1258 1287 1300
##  367 1396 2656  957  661 1642 2069 2764  351  449 1958 2118 2454 2368 1702 1348
## 1312 1321 1333 1378 1381 1435 1456 1467 1493 1547 1560 1579 1590 1637 1640 1663
##  407  291  437  220   47  941 2444 2590   21 2735 2339 1064 2045 2001  630 2227
## 1670 1672 1714 1719 1720 1750 1755 1799 1804 1810 1817 1838 1839 1862 1878 1913
##  641 1706  233  170 1499 1998 2332   32  505 2396 2243  286 1045  791 1183 2370
## 1927 1944 2008 2044 2095 2110 2147 2165 2175 2195 2210 2228 2231 2236 2251 2259
##  434 1179 1983 1772  297
## 2269 2278 2293 2310 2316
##
## DFBETAS for beta Age_Remodel_bc :
## 2296 1638 2686  629 1782 2881 1995 1915 1760 1944  901  137 2228  807 2443 1560
##   39   42   58   65  100  107  112  116  131  198  280  286  289  298  353  372
## 1294 2027 2043 2755 2103  182 2829 1949  716 1611 2273 1996 1860 2330 1558   16
##  398  419  423  431  436  459  489  522  526  544  548  607  615  664  710  718
##   45 2307 1981 1016 2826 1539  153 2380  288 1831 2828 2242 1014 2633  642  605
##  721  726  751  800  840  865  893  894  899  930  983  993 1014 1032 1043 1047
## 2448 1993  758  894 1767 1554 1427  700  392  175 1396  768 1556 2656  957  661
## 1114 1118 1123 1128 1156 1196 1210 1224 1258 1307 1321 1322 1330 1333 1378 1381
## 2074 1513  307 2764 1507 1322  427  751 1287 1958 2118 1221  785  809 1348 2240
## 1385 1426 1448 1467 1470 1508 1552 1555 1556 1560 1579 1583 1620 1622 1663 1699
##  437 2830 2559 1737  719   21 1393 2339 1729 1064 1956 2641  630 1741 1285 2227
## 1714 1722 1767 1785 1796 1804 1808 1817 1826 1838 1853 1871 1878 1884 1903 1913
## 2164 2237  695 2653  278  170 2721  740 1499 2832 1194 1732 2332  215 1738 1643
## 1917 1920 1948 1961 1987 2044 2075 2081 2095 2102 2123 2129 2147 2153 2173 2204
## 2243  791 2467 1183  434 2117 2408 2105 1347
## 2210 2236 2239 2251 2269 2312 2323 2324 2326
##
## DFBETAS for beta Bedroom_AbvGr_bc :
## 2618 1967 1638 2537 2686  629  138 2743 1635 1782 1915  969 1760  301  728  724
##    3   15   42   43   58   65   76   80   92  100  116  124  131  149  183  204
## 2326 1172  901 2228 1369 2677 1537 1332 1780 2443  943 1294 2043  472 2024  182
##  206  219  280  289  301  309  313  317  343  353  370  398  423  442  456  459
##  813 1700 2157 2667  709 2922 1949  716 1181 1363 2003 1000 1860 1188 1947  457
##  464  469  480  511  513  519  522  526  553  579  585  614  615  631  635  650
## 2516 1255   45 2484 2336 2918 1307  666 2648 1180 2380  288  187 1878 2597 1178
##  669  704  721  732  739  757  783  824  861  890  894  899  905  914  974 1023
## 1502  727 2173 2448  758 1071 1767  783  433 1554  955 1169  352 2254 2636 2649
## 1034 1093 1111 1114 1123 1140 1156 1160 1188 1196 1202 1206 1217 1229 1253 1299
##  108  367  768  948  230  661 1303 1513 1642  307 2620 2763 2764 1507 2632  351
## 1301 1312 1322 1335 1366 1381 1425 1426 1435 1448 1458 1462 1467 1470 1473 1493
```

```
## 1322  760 1501  751 2118 2454 2619  785  809 1702 2210 1348   18  291 2294  437
## 1508 1518 1521 1555 1579 1590 1606 1620 1622 1640 1643 1663 1671 1672 1705 1714
##  220 1516  941 1393 1064 2045 2890  630 2286 1360 2651 2237  339 1306 1386  278
## 1719 1736 1750 1808 1838 1839 1849 1878 1879 1899 1901 1920 1936 1955 1978 1987
##  233  454  246 1694 2721  740 1621 1499  353  215  470 2288  285  286 1391 1183
## 2008 2033 2046 2068 2075 2081 2086 2095 2115 2153 2174 2213 2221 2228 2235 2251
## 1387 1179  986 1983 2117
## 2260 2278 2290 2293 2312
##
## DFBETAS for beta Total_Bsmt_SF_bc :
## 1967 2176 2296 1638 2686 1760 1817 1898 1172  897  137 2228  807 2677 1537  679
##   15   30   39   42   58  131  166  180  219  225  286  289  298  309  313  345
## 2493  808 1220  812  182  813  709 2624 1181  795 2003  578 1170 1860 1886 1188
##  362  374  420  449  459  464  513  543  553  565  585  587  608  615  630  631
## 2335  457  815 1267 1255   45 2484  888 1252 2901  677 1539 2380 2698 1878  275
##  649  650  654  674  704  721  732  733  759  786  850  865  894  910  914  924
## 1014 1178 1502  642  605 1557  322 2514 2070 1767  433 1554  392 1342  276  789
## 1014 1023 1034 1043 1047 1077 1088 1132 1135 1156 1188 1196 1258 1276 1298 1300
## 1375  957 2074  307 2620 2764 1507 2632 1501 1221  763 2619 2210  291   84   47
## 1361 1378 1385 1448 1458 1467 1470 1473 1521 1583 1585 1606 1643 1672 1680 1720
## 2559 2590 1498  552 1064 2890 2088  787  630  811 2227  274 2485  170 1968 1499
## 1767 1799 1832 1835 1838 1849 1851 1876 1878 1898 1913 1986 2036 2044 2088 2095
##  650  953  791 1183 1504 1983  155 1772  297  623
## 2167 2219 2236 2251 2287 2293 2296 2310 2316 2318
##
## DFBETAS for beta Garage_Cars_bc :
## 2618 1967 1311  557 2880 1987  753  728 1515 1132  901 2677 1332 2493  943  808
##    3   15   51   72  129  139  140  183  227  279  280  309  317  362  370  374
##  812  182  813 1942 2667 2193 1547  204 1319 1806 1947  292 2747 1558 2839 1950
##  449  459  464  479  511  530  551  558  575  604  635  657  676  710  875  896
##  120  187 1505  536 1502  214  605 1296  713 2126  727 2173 1993  758  894  853
##  901  905  920  932 1034 1041 1047 1066 1078 1089 1093 1111 1118 1123 1128 1141
## 2872 1767  783 2879  898 1554 2668 2254 1412  276  131 2656  721 2687  984  307
## 1155 1156 1160 1170 1187 1196 1203 1229 1285 1298 1319 1333 1365 1422 1440 1448
## 2620 1507 2632   28 1322  760 1501  325 2834 2118 1221  763 2232 2619 1600  393
## 1458 1470 1473 1492 1508 1518 1521 1540 1578 1579 1583 1585 1593 1606 1629 1639
## 2210  291 2240 2294 1516  541 1376 2045  630  811 1285 1514 1306 1386  278  233
## 1643 1672 1699 1705 1736 1745 1823 1839 1878 1898 1903 1933 1955 1978 1987 2008
## 2914  170 1550 1499  286 1504 2192
## 2017 2044 2080 2095 2228 2287 2295
##
## DFBETAS for beta Full_Bath_bc :
## 2296 1638 2686 2743 2852 1944  724  807 1537 1232  766 1294 1025 2043 2352  337
##   39   42   58   80  197  198  204  298  313  373  390  398  405  423  424  450
##  716 1181 1996 1170 2664 2840 1016 1539 1502  727 1281 2514 2860 1554 2636  789
##  526  553  607  608  693  760  800  865 1034 1093 1124 1132 1191 1196 1253 1300
##  768 1556 2656  957 1303 1513 2764 1501  751 2118  763  809 1348  291 2294  437
## 1322 1330 1333 1378 1425 1426 1467 1521 1555 1579 1585 1622 1663 1672 1705 1714
##  220 1646  719 1064 2045 1314 2890 2001 2227 1386   78  233  170 2721 1499 2832
## 1719 1775 1796 1838 1839 1842 1849 1862 1913 1978 1985 2008 2044 2075 2095 2102
## 2332  285  286  791 1504
## 2147 2221 2228 2236 2287
##
## DFBETAS for beta Fireplaces_bc :
```

```
## 2296 2686  629 2743 1336 1995 1760  109  724  838  593 1537 2443  943 1560 1294
##   39   58   65   80   95  112  131  203  204  238  312  313  353  370  372  398
## 2043 1325  182  158 2099  716 1611 1181 1978 2172 1996 1170  457 2516   16 1981
##  423  434  459  505  509  526  544  553  569  606  607  608  650  669  718  751
## 2840 1567 2839 2844 2828 2204 2219 1014  642  605 2448 1993 2514  967 1767  783
##  760  834  875  979  983  984 1002 1014 1043 1047 1114 1118 1132 1153 1156 1160
##  955 2668  352  700 2039  951  859 2636 1556 2656  721 2074 2693 1401 1976  307
## 1202 1203 1217 1224 1225 1233 1242 1253 1330 1333 1365 1385 1405 1427 1428 1448
## 2069 2764   28 1322  785 1529 2210  220 2830  680  941 2559  719  552 1064 2890
## 1456 1467 1492 1508 1620 1632 1643 1719 1722 1741 1750 1767 1796 1835 1838 1849
## 1956 2001 2814  630 1741 2164 2237 1386   78  278  170  280  740 1621 1499 2832
## 1853 1862 1873 1878 1884 1917 1920 1978 1985 1987 2044 2063 2081 2086 2095 2102
## 1998  353  286  791 1571 1183 2370 1179 1983 2117 2408 1347
## 2110 2115 2228 2236 2247 2251 2259 2278 2293 2312 2323 2326
##
## DFBETAS for beta Bldg_Type2Non1Fam :
## 2642 2176  689  284 1782 2881 1995 1760  299 1172   49 1632 2677 2364 2443 1560
##   21   30   73   94  100  107  112  131  199  219  266  282  309  339  353  372
## 2027  337  182  813 2829  709 2193  219 1181  190 1000  409 2664  976 1313   16
##  419  450  459  464  489  513  530  536  553  559  614  675  693  702  709  718
## 2336 2918  666 2826 1483 1044  187 1168 1505  295 1634 2287 2828 1861 1178  642
##  739  757  824  840  854  868  905  918  920  945  950  972  983  987 1023 1043
##  605  727 1153 2173 2448 1993  758 1071 1767 2517 1554  700 2800 2254 2156 1412
## 1047 1093 1109 1111 1114 1118 1123 1140 1156 1172 1196 1224 1227 1229 1230 1285
## 1907  276 2649 2656 1303 1597 2632  351   35 2118  763  817  818 2368 1484  407
## 1287 1298 1299 1333 1425 1442 1473 1493 1496 1579 1585 1624 1631 1637 1669 1670
##  291  816 2240 2294  437 2830  941 2559 2590  136  552 2890 2286 1285 2908  695
## 1672 1674 1699 1705 1714 1722 1750 1767 1799 1800 1835 1849 1879 1903 1914 1948
##  418  978   78  233 2914   76  569 1499 1015 1998  353 2602  215 2912   32  470
## 1963 1974 1985 2008 2017 2054 2059 2095 2096 2110 2115 2138 2153 2164 2165 2174
##  107  327 2243 2288  286 1045 1640 1571 2370 1387 1485 2546 1983 2192
## 2180 2208 2210 2213 2228 2231 2240 2247 2259 2260 2276 2291 2293 2295
##
## DFBETAS for beta Gr_Liv_Area_bc:Bldg_Type2Non1Fam :
## 2642 2176  689  284 2881 1995 1760 1817  469 1172   49 1632 2364 2443 1560 2027
##   21   30   73   94  107  112  131  166  193  219  266  282  339  353  372  419
##  337  182  813 2829  709 2193 1181  190 1000 2330  409 2664  976 1313   16 2336
##  450  459  464  489  513  530  553  559  614  664  675  693  702  709  718  739
## 2918  666 2826 1483 1044  187 1168 1505  295 2287 2828 1861 1178  642  605  727
##  757  824  840  854  868  905  918  920  945  972  983  987 1023 1043 1047 1093
## 1153 2173 2448 1993  758 1071 1767 2517 1554  700 2800 2254 2156 1412 1907  276
## 1109 1111 1114 1118 1123 1140 1156 1172 1196 1224 1227 1229 1230 1285 1287 1298
## 2649 2656 1303 1597 2632  351   35  325 2118  763  817  818 2368  407  291  816
## 1299 1333 1425 1442 1473 1493 1496 1540 1579 1585 1624 1631 1637 1670 1672 1674
## 2240 2294  437 2830  941 2559 2590  136  552 2286 2227 2908 2888  695  418  978
## 1699 1705 1714 1722 1750 1767 1799 1800 1835 1879 1913 1914 1943 1948 1963 1974
##   78  233 2914  569 1621 1499 1015 1998  353 2602  215 2912   32  470  107 2243
## 1985 2008 2017 2059 2086 2095 2096 2110 2115 2138 2153 2164 2165 2174 2180 2210
## 2288  953 1045 1640 1571 2370 1387  434 1485 2546 1983 2192
## 2213 2219 2231 2240 2247 2259 2260 2269 2276 2291 2293 2295
```

```
## find how many influential points are in at least 3 categories
is_lev    <- h_ii > h_cut
is_out    <- (r_std > 4 | r_std < -4)
```

```r
is_cook   <- cook_d > cook_cut
is_dffits <- abs(dffits_i) > fit_cut

flag_count <- is_lev + is_out + is_cook + is_dffits

idx_3plus <- which(flag_count >= 3)
idx_3plus
```

```
## 1554 2890 1499 1183
## 1196 1849 2095 2251
```

```r
# find their dfbeta count
dfb_flag <- abs(dfbetas_i) > beta_cut
dfb_count <- rowSums(dfb_flag)


# summary table
infl_summary <- data.frame(
  id         = idx_3plus,
  n_flags    = flag_count[idx_3plus],
  n_badbetas = dfb_count[idx_3plus],
  leverage   = is_lev[idx_3plus],
  outlier    = is_out[idx_3plus],
  cook       = is_cook[idx_3plus],
  dffits     = is_dffits[idx_3plus]
)
infl_summary
```

```
##        id n_flags n_badbetas leverage outlier  cook dffits
## 1554 1196       3         10     TRUE    TRUE FALSE   TRUE
## 2890 1849       3          6     TRUE    TRUE FALSE   TRUE
## 1499 2095       3         11     TRUE    TRUE FALSE   TRUE
## 1183 2251       3          7     TRUE    TRUE FALSE   TRUE
```

test

```r
# do the same transformation on test

test$Gr_Liv_Area_bc    <- sqrt(test$Gr_Liv_Area)
test$Total_Bsmt_SF_bc <- sqrt(test$Total_Bsmt_SF)
test$Mas_Vnr_Area_bc   <- sqrt(test$Mas_Vnr_Area)

test$Age_Remodel_bc    <- log(test$Age_Remodel + 1)
test$Bedroom_AbvGr_bc <- log(test$Bedroom_AbvGr + 1)

test$Garage_Cars_bc    <- (test$Garage_Cars)^(1/3)
test$Full_Bath_bc      <- (test$Full_Bath)^(1/4)
test$Fireplaces_bc     <- (test$Fireplaces)^(1/4)

test$Y_bc <- bc_transform(test$Sale_Price, lambda_y)
```

```r
summary(test[, c("Y_bc", "Gr_Liv_Area_bc","Total_Bsmt_SF_bc","Mas_Vnr_Area_bc",
                 "Age_Remodel_bc","Bedroom_AbvGr_bc","Garage_Cars_bc",
                 "Full_Bath_bc","Fireplaces_bc")])
```

```
##       Y_bc         Gr_Liv_Area_bc  Total_Bsmt_SF_bc Mas_Vnr_Area_bc
##  Min.   : 50.71   Min.   :21.91   Min.   : 0.00    Min.   : 0.000
##  1st Qu.: 71.51   1st Qu.:33.19   1st Qu.:27.93    1st Qu.: 0.000
##  Median : 75.37   Median :37.52   Median :31.14    Median : 0.000
##  Mean   : 76.69   Mean   :37.67   Mean   :31.02    Mean   : 5.528
##  3rd Qu.: 81.32   3rd Qu.:41.35   3rd Qu.:35.45    3rd Qu.:12.073
##  Max.   :105.18   Max.   :58.22   Max.   :55.62    Max.   :33.392
##  Age_Remodel_bc  Bedroom_AbvGr_bc Garage_Cars_bc   Full_Bath_bc
##  Min.   :0.000   Min.   :0.000    Min.   :0.000   Min.   :0.000
##  1st Qu.:1.792   1st Qu.:1.099    1st Qu.:1.000   1st Qu.:1.000
##  Median :2.862   Median :1.386    Median :1.260   Median :1.189
##  Mean   :2.652   Mean   :1.311    Mean   :1.142   Mean   :1.091
##  3rd Qu.:3.807   3rd Qu.:1.386    3rd Qu.:1.260   3rd Qu.:1.189
##  Max.   :4.111   Max.   :1.946    Max.   :1.587   Max.   :1.316
##  Fireplaces_bc
##  Min.   :0.0000
##  1st Qu.:0.0000
##  Median :0.0000
##  Mean   :0.5137
##  3rd Qu.:1.0000
##  Max.   :1.3161
```

```r
# prediction of train set
pred_train_bc <- predict(final_bc, newdata = train)

# predicttion of test set
pred_test_bc  <- predict(final_bc, newdata = test)

# MSE
mse_train_bc <- mean((train$Y_bc - pred_train_bc)^2)
mse_test_bc  <- mean((test$Y_bc  - pred_test_bc)^2)

mse_train_bc
```

```
## [1] 12.95178
```

```r
mse_test_bc
```

```
## [1] 12.96401
```

```r
bc_inverse <- function(y_bc, lambda = 0.25) {
  (lambda * y_bc + 1)^ (1 / lambda)
}

# 拉回原始房价
pred_train_price <- bc_inverse(pred_train_bc, lambda = 0.25)
```

```r
pred_test_price  <- bc_inverse(pred_test_bc,  lambda = 0.25)

mse_train_price  <- mean((train$Sale_Price - pred_train_price)^2)
mse_test_price   <- mean((test$Sale_Price  - pred_test_price)^2)

rmse_train_price <- sqrt(mse_train_price)
rmse_test_price  <- sqrt(mse_test_price)

rmse_train_price
```

```
## [1] 36189.55
```

```r
rmse_test_price
```

```
## [1] 31527.88
```

```r
rmse_train_bc <- sqrt(mse_train_bc)
rmse_test_bc  <- sqrt(mse_test_bc)
sd_ybc        <- sd(train$Y_bc)

rmse_train_bc
```

```
## [1] 3.598858
```

```r
rmse_test_bc
```

```
## [1] 3.600557
```

```r
sd_ybc
```

```
## [1] 8.367965
```

```r
rmse_train_bc / sd_ybc
```

```
## [1] 0.4300757
```

```r
rmse_test_bc  / sd_ybc
```

```
## [1] 0.4302787
```

On the Box–Cox–transformed scale, the training and test MSEs are 12.95 and 12.96 (RMSE   3.60), which correspond to about 43% of the standard deviation of the transformed response. This indicates that the model explains a meaningful portion of the variation in sale prices, but there is still substantial unexplained variability.

On the original price scale, the training and test RMSEs are approximately $36,000 and $31,500, respectively. The fact that the test error is very close to the training error suggests that the model does not overfit severely and generalizes reasonably well to new data, although the prediction errors remain non-trivial in economic terms.

```r
# training model: 已经有了
mod_train <- final_bc   # 就是 step_bc

# test 上用完全一样的公式 fit 一次（只是为了算诊断量，不用于选模型）
mod_test <- lm(
  Y_bc ~ Gr_Liv_Area_bc + Mas_Vnr_Area_bc + Age_Remodel_bc +
    Bedroom_AbvGr_bc + Total_Bsmt_SF_bc + Garage_Cars_bc +
    Full_Bath_bc + Fireplaces_bc + Bldg_Type2 +
    Gr_Liv_Area_bc:Bldg_Type2,
  data = test
)

p      <- length(coef(mod_train)) - 1   # 10 个 predictor
n_tr   <- nrow(train)
n_te   <- nrow(test)

# cutoff 都用你现在这一套:
hcut_tr  <- 2 * (p + 1) / n_tr
hcut_te  <- 2 * (p + 1) / n_te

cookcut_tr <- qf(0.5, df1 = p + 1, df2 = n_tr - p - 1)
cookcut_te <- qf(0.5, df1 = p + 1, df2 = n_te - p - 1)

fitcut_tr  <- 2 * sqrt((p + 1) / n_tr)
fitcut_te  <- 2 * sqrt((p + 1) / n_te)


# training set
n_lev_tr    <- length(which(hatvalues(mod_train) > hcut_tr))
n_out_tr    <- length(which(abs(rstandard(mod_train)) > 4))
n_cook_tr   <- length(which(cooks.distance(mod_train) > cookcut_tr))
n_dffits_tr<- length(which(abs(dffits(mod_train)) > fitcut_tr))

cbind(n_lev_tr, n_out_tr, n_cook_tr, n_dffits_tr)
```

```
##      n_lev_tr n_out_tr n_cook_tr n_dffits_tr
## [1,]      222        9         0         142
```

```r
# test set
n_lev_te    <- length(which(hatvalues(mod_test) > hcut_te))
n_out_te    <- length(which(abs(rstandard(mod_test)) > 4))
n_cook_te   <- length(which(cooks.distance(mod_test) > cookcut_te))
n_dffits_te<- length(which(abs(dffits(mod_test)) > fitcut_te))

cbind(n_lev_te, n_out_te, n_cook_te, n_dffits_te)
```

```
##      n_lev_te n_out_te n_cook_te n_dffits_te
## [1,]       53        1         0          52
```

When we refit the final model on the test set and counted high-leverage points, extreme standardized residuals and large DFFITS using the same cutoffs, the proportions of flagged observations were similar in the training and test sets (e.g., 222 vs 53 high-leverage points and 142 vs 52 large-DFFITS points under an 80/20 split). This suggests that potentially problematic observations are not unique to one split of the data, although our main influential-point analysis is based on the training sample.